

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**End Semester Examination – Winter 2018**

**Course: B. Tech in Mechanical Engineering**

**Sem: III**

**Subject Name: Fluid Mechanics**

**Subject Code: BTMEC303**

**Max Marks:60**

**Date: 05-12-2018**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                                               | (Level/CO)          | Marks      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------|
| <b>Q. 1 Solve Any <u>Two</u> of the following.</b>                                                                                                                                                                                                                                                                                                                                                                            |                     |            |
| A) State and derive Pascal's law.                                                                                                                                                                                                                                                                                                                                                                                             | <b>(Remember)</b>   | <b>(6)</b> |
| B) A cubical block weighing 4.5 N and having a 40 cm edge is allowed to slide down an inclined plane surface making an angle of $30^\circ$ with the horizontal on which there is a uniform layer of oil 0.005 cm thick. If the expected steady state velocity of the block is 12.5 cm/s, determine viscosity of the oil. Also express the kinematic viscosity in stokes if the oil has mass density of $800 \text{ kg/m}^3$ . | <b>(Apply)</b>      | <b>(6)</b> |
| C) List some occurrences which can be attributed to the physics of surface tension. Set-up a relationship between surface tension and pressure intensity (in excess of outside pressure) for a liquid droplet.                                                                                                                                                                                                                | <b>(Analysis)</b>   | <b>(6)</b> |
| <b>Q.2 Solve the following questions.</b>                                                                                                                                                                                                                                                                                                                                                                                     |                     |            |
| A) A rectangular plate 3 m x 5m is immersed vertically in water such that the 3 m side is parallel to the water surface. Determine the hydrostatic force and centre of pressure if the top surface is i) flush with the water surface ii) 2 m below the water surface.                                                                                                                                                        | <b>(Apply)</b>      | <b>(6)</b> |
| B) A solid cylinder of diameter 4.0m has a height of 3m .Find the meta-centric height of the cylinder when it is floating in water with its axis vertical. The specific gravity of the cylinder = 0.6. Comment on the stability of cylinder                                                                                                                                                                                   | <b>(Understand)</b> | <b>(6)</b> |
| <b>Q. 3 Solve the following questions.</b>                                                                                                                                                                                                                                                                                                                                                                                    |                     |            |
| A) Derive the differential form of continuity equation in Cartesian coordinate system.                                                                                                                                                                                                                                                                                                                                        | <b>(Understand)</b> | <b>(6)</b> |
| B) A stream function is given as $\psi = 2xy$ . Determine i) whether the flow is possible ii) whether the flow is rotational or irrotational                                                                                                                                                                                                                                                                                  | <b>(Analysis)</b>   | <b>(6)</b> |

**Q.4 Solve the following questions.**

- A) State Bernoulli's theorem. Derive Euler's equation of motion and from it derive Bernoulli's equation. (Understand) (6)
- B) A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm mercury. Determine the rate of flow. Take  $C_d=0.98$  (Apply) (6)

**Q. 5 Solve the following questions.**

- A) Derive an expression for the velocity distribution for viscous flow through a circular pipe. (Understand) (6)
- B) Determine the difference in the elevations between the water surfaces in the two tanks which are connected by a horizontal pipe of diameter 300 mm and length 400 m. The rate of flow of water through the pipe is 300 litres/s. Consider all losses and take the value of  $f = 0.008$  (Analysis) (6)

**Q. 6 Solve Any Two of the following questions.**

- A) In a compressor, the frictional torque  $T$  in the impeller depends on diameter  $D$ , rotational speed  $N$ , fluid density  $\rho$  and viscosity  $\mu$ . Using Buckingham's  $\pi$  theorem, obtain an expression for torque. (Apply) (6)
- B) The velocity distribution in the boundary layer is given by  $\frac{u}{U} = \left(\frac{y}{\delta}\right)^{\frac{1}{7}}$ . Calculate the displacement thickness and momentum thickness. (Apply) (6)
- C) What do you understand by the terms drag, lift and coefficient of drag? (Recall) (6)

\*\*\* End \*\*\*

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****End Semester Examination – Summer 2019****Course: B. Tech in Mechanical Engineering****Sem: III****Subject Name: Fluid Mechanics****Subject Code: BTMEC303****Max Marks: 60****Date: 30/05/2019****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                           | (Level/CO) | Marks    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| <b>Q. 1 A)</b> Explain with the neat sketch U-tube differential Manometer.                                                                                                                                                                                                                                                                                                                                |            | <b>6</b> |
| <b>B)</b> A vertical gap 2.2cm wide of infinite extent contains a fluid of viscosity $2.0\text{Ns/m}^2$ and specific gravity 0.9. A metallic plate $1.2\text{m} \times 1.2\text{m} \times 0.2\text{cm}$ is to be lifted up with a constant velocity of $0.15\text{m/sec}$ , through the gap. If the plate is in the middle of the gap, find the force required. The weight of the plate is $40\text{N}$ . |            | <b>6</b> |
| <b>Q.2 A)</b> Explain with the neat sketch the condition for equilibrium for floating bodies.                                                                                                                                                                                                                                                                                                             |            | <b>6</b> |
| <b>B)</b> A rectangular plane surface is $2\text{m}$ wide and $3\text{m}$ deep. It lies in vertical plane to water. Determine the total pressure and position of centre of pressure on the plane surface when its upper edge is horizontal and a) coincides with water surface, b) $2.5\text{m}$ below the free water surface.                                                                            |            | <b>6</b> |
| <b>Q. 3 A)</b> Derive the general equation for continuity for a three dimensional flow in Cartesian Co-ordinates for a steady incompressible flow.                                                                                                                                                                                                                                                        |            | <b>6</b> |
| <b>B)</b> Explain the various types of fluid flows.                                                                                                                                                                                                                                                                                                                                                       |            | <b>6</b> |
| <b>Q.4 A)</b> Derive the expression for discharge over a triangular notch.                                                                                                                                                                                                                                                                                                                                |            | <b>6</b> |
| <b>B)</b> Describe an Venturimeter.                                                                                                                                                                                                                                                                                                                                                                       |            | <b>6</b> |
| <b>Q. 5 A)</b> Prove that the maximum velocity in a circular pipe for viscous flow is equal to two times the average velocity of the flow.                                                                                                                                                                                                                                                                |            | <b>6</b> |
| <b>B)</b> What is minor energy loss in the pipe? Explain various types of minor losses in pipe flow.                                                                                                                                                                                                                                                                                                      |            | <b>6</b> |
| <b>Q. 6 A)</b> State Buckingham's $\pi$ -Theorem and explain procedure for determining the $\pi$ -groups and their functional relationship.                                                                                                                                                                                                                                                               |            | <b>6</b> |
| <b>B)</b> What are the different methods of preventing the separation of boundary layers. Explain with the neat sketches.                                                                                                                                                                                                                                                                                 |            | <b>6</b> |

**\*\*\* End \*\*\***

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –**  
**RAIGAD -402 103**  
**Winter Semester Examination – December - 2019**

**Branch: Mechanical Engineering**

**Sem.: -III**

**Subject with Subject Code:-Fluid Mechanics - BTMEC303**

**Marks:60**

**Date:-14/12/2019**

**Time:- 3 Hr.**

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**Instructions to the Students**

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

**(Marks)**  
**(06)**

**Q.1. (a) Define the following fluid properties:**

- (i) Viscosity
- (ii) Compressibility
- (iii) Surface Tension

(b) A U-tube manometer is used to measure the pressure of water in a pipe line, which is excess of atmospheric pressure. The right limb of the manometer contains mercury and open to atmosphere. The contact between water and mercury is in the left limb. Determine the pressure of the water in the main line. If the difference in the level of mercury in the limbs of U-tube is 10cm and the free surface of mercury is in the level with the center of the pipe. If the pressure of water in the pipe line reduced to  $9810 \text{ N/m}^2$ . Calculate the new difference in the level of mercury. Sketch the arrangement in both cases.

**(06)**

Q.2.(a) Prove that the Center of Pressure of a completely sub-merged plane surface is always below the Center of Gravity of the sub-merged surface when the plane surface is vertical. (06)

(b) What are the conditions of equilibrium of floating body and sub-merged body? (06)

Q.3.(a) Define the following flow: (06)

- (i) Steady Flow
- (ii) Non-Uniform Flow
- (iii) Laminar Flow
- (iv) Turbulant Flow
- (v) Compressible Flow
- (vi) Irrotational Flow

(b) Derive an expression of three dimensional continuity equation in rectangular coordinate system. (06)

**OR**

(b) If for a two-dimensional potential flow, the velocity potential is given by  $\Phi = x(2y - 1)$ . Determine the velocity at the point P(4,5) and value of stream function at the point P. (06)

Q.4.(a) Derive an expression for the Discharge through Triangular Notch. (06)

(b) A 30cm×15cm venturimeter is provided in a vertical pipe line carrying oil of specific gravity 0.9, the flow being upward. The difference in elevation of the throat section and entrance section of the venturimeter is 30cm. The

differential U-tube mercury manometer shows a difference of mercury level 25cm.

Calculate:

(i) The discharge of oil

(ii) The pressure difference between entrance section and throat section.

Take  $C_d = 0.98$  and specific gravity of mercury = 13.6 (06)

Q.5.(a) Derive an expression for shear stress distribution and velocity distribution for Laminar flow through circular pipe. (06)

(b) A horizontal pipe line 40m long is connected to a water tank at one end and discharges freely into atmosphere at other end. For the first 25m of its length from the tank, the pipe is 150mm diameter and its diameter is suddenly enlarged to 300mm. The height of water level in the tank is 8m above the center of the pipe. Considering all losses of head which occur. Determine the rate of flow. Take  $f = 0.01$  for both pipe. (06)

Q.6.(a) The efficiency  $\eta$  of fan depends on density  $\rho$ , dynamic viscosity  $\mu$  of the fluid, angular velocity  $\omega$ , diameter  $D$  of the rotor and the discharge  $Q$ . Express efficiency  $\eta$  of fan in terms of dimensionless parameters. (06)

(b) Define Displacement thickness. Derive an expression for displacement thickness. (06)

**OR**

(b) What do you understand by: Total drag on the body, Resultant force on a body, co-efficient of drag and co-efficient of lift. (06)

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Winter Examination – 2022**

**Course: B. Tech.      Branch: Mechanical Engg.      Semester : III**

**Subject Code & Name: BTMC302 / Fluid Mechanics**

**Max Marks: 60**

**Date: 11 Mar 2023**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (Level/CO)           | Marks     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| <b>Q. 1 Attempt the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      | <b>12</b> |
| <b>A)</b> Describe any three terms:<br>1) Density, 2) Pascal's Law, 3) Specific Gravity,<br>4) Surface Tension, 5) Mach number and its significance                                                                                                                                                                                                                                                                                                                   | Understand           | <b>6</b>  |
| <b>B)</b> i) Calculate the capillary rise in a glass tube of 2.0 mm diameter when immersed vertically in (a) water and (b) Mercury. Take surface tension = 0.08 N/m for water and 0.5 N/m for mercury in contact with air. The specific gravity of mercury is 13.6 and angle of contact is 120°.<br><br>ii) A hydraulic press has a ram of 20 cm diameter and plunger of 3 cm diameter. It is used to lift a weight of 30 kN. Find the force required at the plunger. | Evaluate             | <b>6</b>  |
| <b>C)</b> What is the purpose of a manometer? Describe the various types of manometers along with neat schematics and its working.                                                                                                                                                                                                                                                                                                                                    | Understand           | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |                      | <b>12</b> |
| <b>A)</b> A rectangular vertical plane surface is 2 m wide and 3 m deep. Determine the total pressure force and position of centre of pressure on the plane surface when its upper edge is horizontal and coincides with water level. Also determine the force and centre of pressure position when the top edge of plane surface is 2.5 m below the water level.                                                                                                     | Evaluate             | <b>6</b>  |
| <b>B)</b> Describe the conditions of stability of floating as well as submerged bodies along with neat schematic and tabular presentation.                                                                                                                                                                                                                                                                                                                            | Understand           | <b>6</b>  |
| <b>C)</b> A rectangular block is 5.0 m long, 3 m wide and 1.2 m high. The depth of immersion of the block is 0.8 m in the sea water. If the centre of gravity is 0.6 m above the bottom of the block, determine the Metacentric height. Assume density of sea water as 1025 kg/cu. m.                                                                                                                                                                                 | Analyze and Evaluate | <b>6</b>  |
| <b>Q. 3 Attempt any two of the following</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                      | <b>12</b> |
| <b>A)</b> Derive the Bernoulli's equation with the help of Euler's equation. Also discuss the significance of each terms of Bernoulli's equation.                                                                                                                                                                                                                                                                                                                     | Understand           | <b>6</b>  |
| <b>B)</b> An oil of viscosity 0.1 Ns/m <sup>2</sup> and relative density 0.9 is flowing through a circular pipe of diameter 50 mm and of length 300 m. The rate of flow of fluid through the pipe is 3.5 litres/s. Find the pressure drop in a length of 300 mm and also the shear stress at the pipe wall.                                                                                                                                                           | Analyze and Evaluate | <b>6</b>  |
| <b>C)</b> A 0.30 m × 0.15 m Venturimeter is inserted in a vertical pipe of diameter 0.30 m. The throat size is 0.15 m. The water is flowing through the pipe in the upward direction. A differential mercury manometer connected to the inlet and throat of Venturimeter gives a reading of 0.2 m. Find the flow rate of water (Assume, C <sub>d</sub> = 0.98).                                                                                                       | Evaluate             | <b>6</b>  |

|                                                                                                         |            |           |
|---------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q.4 Attempt any two of the following</b>                                                             |            | <b>12</b> |
| <b>A)</b> Describe in detail the various types of major and minor losses in pipes.                      | Understand | <b>6</b>  |
| <b>B)</b> Explain: i) Dimensional Homogeneity and ii) Raleigh's method                                  | Understand | <b>6</b>  |
| <b>C)</b> Explain: i) Water hammer effect in pipes, ii) Hydraulic gradient line and Total gradient line | Remember   | <b>6</b>  |

|                                                                                                                                                                                                                                                      |                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| <b>Q. 5 Attempt the following</b>                                                                                                                                                                                                                    |                      | <b>12</b> |
| <b>A)</b> Describe Reynolds experiment with the help of neat sketches along with the significance of Reynolds number.                                                                                                                                | Understand           | <b>6</b>  |
| <b>B)</b> Three pipes of lengths 800m, 500m, and 400 m and of diameters 50 cm, 40 cm and 30 cm, respectively are connected in series. These pipes are to be replaced by a single pipe of length 1700 m. Find the diameter of single equivalent pipe. | Analyze and Evaluate | <b>6</b>  |

\*\*\* End \*\*\*

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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary End Semester Examination – Winter 2022-23</b><br><b>Course: B. Tech.      Branch: Mechanical Engg.      Semester : III</b><br><b>Subject Code &amp; Name: Fluid Mechanics (BTMEC303)</b><br><b>Max Marks: 60      Date: __/01/2023      Duration: 3 Hr.</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      |              |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      |              |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Describe the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>(Level/CO)</b>    | <b>Marks</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe any three terms:<br>1) Mach number, 2) Pascal's Law, 3) Manometer,<br>4) Steady and unsteady flow, 5) Incompressible fluid flow                                                                                                                                                                                                                                                                                                                    | Understand           | <b>6</b>     |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | i) Calculate the capillary rise in a glass tube of 2.0 mm diameter when immersed vertically in (a) water and (b) Mercury. Take surface tension = 0.08 N/m for water and 0.5 N/m for mercury in contact with air. The specific gravity of mercury is 13.6 and angle of contact is 120°.<br><br>ii) A hydraulic press has a ram of 25 cm diameter and plunger of 4 cm diameter. It is used to lift a weight of 50 kN. Find the force required at the plunger. | Evaluate             | <b>3 × 2</b> |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Attempt any two questions A/B/C</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |              |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | A rectangular vertical plane surface is 2 m wide and 2 m deep. Determine the total pressure force and position of centre of pressure on the plane surface when its upper edge is horizontal and coincides with water level. Also determine the force and centre of pressure position when top edge of plane surface is 1.5 m below the water level.                                                                                                         | Evaluate             | <b>6</b>     |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe the conditions of stability of floating as well as submerged bodies along with neat schematic and tabular presentation.                                                                                                                                                                                                                                                                                                                            | Understand           | <b>6</b>     |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | A rectangular block is 5 m long, 4 m wide and 1.4 m high. The depth of immersion of the block is 0.8 m in the sea water. If the centre of gravity is 0.7 m above the bottom of the block, determine the metacentric height. Assume density of sea water as 1020 kg/cu. m.                                                                                                                                                                                   | Analyze and Evaluate | <b>6</b>     |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Attempt any two of the following</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |                      |              |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | Derive the Bernoulli's equation with the help of Euler's equation.                                                                                                                                                                                                                                                                                                                                                                                          | Understand           | <b>6</b>     |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | The right limb of a simple U-tube manometer containing mercury is open to the atmosphere while the left limb is connected to a pipe in which a fluid of specific gravity 0.85 is flowing. The centre of the pipe is 0.12 m below the level of mercury in the right limb. Find the pressure of fluid in the pipe if the difference of mercury level in the two limbs is 0.2 m.                                                                               | Analyze and Evaluate | <b>6</b>     |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | A 0.35 m × 0.18 m Venturimeter is inserted in a vertical pipe of diameter 0.35 m. The throat size is 0.18 m. The water is flowing through the pipe in the upward direction. A differential                                                                                                                                                                                                                                                                  | Evaluate             | <b>6</b>     |

|             |                                                                                                                                                                                                                                           |                      |          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|
|             | mercury manometer connected to the inlet and throat of Venturimeter gives a reading of 0.2 m. Find the flow rate of water (Assume, $C_d = 0.98$ ).                                                                                        |                      |          |
|             |                                                                                                                                                                                                                                           |                      |          |
| <b>Q.4</b>  | <b>Attempt any two of the following</b>                                                                                                                                                                                                   |                      |          |
| A)          | Describe in detail the various types of major and minor losses in pipes.                                                                                                                                                                  | Understand           | <b>6</b> |
| B)          | Explain: i) Dimensional Homogeneity and ii) Raleigh's method                                                                                                                                                                              | Understand           | <b>6</b> |
| C)          | Explain: i) Water hammer effect in pipes, ii) Hydraulic gradient line and Total gradient line                                                                                                                                             | Remember             | <b>6</b> |
|             |                                                                                                                                                                                                                                           |                      |          |
| <b>Q. 5</b> | <b>Attempt the following</b>                                                                                                                                                                                                              |                      |          |
| A)          | Describe Reynolds experiment with the help of neat sketches along with the significance of Reynolds number.                                                                                                                               | Understand           | <b>6</b> |
| B)          | Three pipes of lengths 800m, 600m, and 400 m and of diameters 40 cm, 35cm and 30 cm, respectively are connected in series. These pipes are to be replaced by a single pipe of length 1600 m. Find the diameter of single equivalent pipe. | Analyze and Evaluate | <b>6</b> |
|             | <b>*** End ***</b>                                                                                                                                                                                                                        |                      |          |

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Course: B. Tech.

Branch : Mechanical

Semester : III

Subject Code & Name: BTMC302 , Fluid Mechanics

Max Marks: 60

Date: 10/08/2023

Duration: 3 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                          | (Level/CO) | Marks     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                              |            | <b>12</b> |
| <b>A)</b> What are the conditions of equilibrium of floating body and sub-merged Body?                                                                                                                                                                                                                                                                                                                   | <b>CO1</b> | <b>6</b>  |
| <b>B)</b> Define and explain the following fluid properties.<br>Viscosity, specific gravity, specific volume, specific weight, surface tension, compressibility.                                                                                                                                                                                                                                         | <b>CO1</b> | <b>6</b>  |
| <b>C)</b> The space between two square flat parallel plates is filled with oil. Each side of the plate is 60 cm. The thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 metre per sec requires a force of 98.1 N to maintain the speed. Determine : (i) the dynamic viscosity of the oil , and (ii) the kinematic viscosity of the oil if the specific gravity of the oil is 0.95 | <b>CO1</b> | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                               |            | <b>12</b> |
| <b>A)</b> Define the following flow:<br>(i) Steady Flow<br>(ii) Non-Uniform Flow<br>(iii) Laminar Flow<br>(iv) Turbulent Flow<br>(v) Compressible Flow<br>(vi) Irrotational Flow                                                                                                                                                                                                                         | <b>CO2</b> | <b>6</b>  |
| <b>B)</b> Derive the general equation for continuity for a three dimensional flow in Cartesian Co-ordinates for a steady incompressible flow                                                                                                                                                                                                                                                             | <b>CO2</b> | <b>6</b>  |
| <b>C)</b> The velocity potential function is given by $\phi = 5(x^2 - y^2)$ . Calculate the velocity components at the point (4, 5).                                                                                                                                                                                                                                                                     | <b>CO2</b> | <b>6</b>  |

|             |                                                                                                                                                                                                                                     |            |           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                              |            | <b>12</b> |
| <b>A)</b>   | Derive an expression for the velocity distribution for laminar flow through a circular pipe. Also sketch the velocity distribution and shear stress distribution across a section of the pipe                                       | <b>CO3</b> | <b>6</b>  |
| <b>B)</b>   | What is minor energy loss in the pipe? Explain various types of minor losses in pipe flow.                                                                                                                                          | <b>CO3</b> | <b>6</b>  |
| <b>C)</b>   | Calculate the discharge through a pipe of diameter 200 mm when the difference of pressure head between the two ends of a pipe 500 m apart is 4 m of water. Take the value of $f = 0.009$ in the formula $h_f = \frac{4fLV^2}{2g*d}$ | <b>CO3</b> | <b>6</b>  |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                              |            | <b>12</b> |
| <b>A)</b>   | State Buckingham's $\pi$ -Theorem and explain procedure for determining the $\pi$ -groups and their functional relationship                                                                                                         | <b>CO4</b> | <b>6</b>  |
| <b>B)</b>   | What is Siphon? Explain its working with neat sketch.                                                                                                                                                                               | <b>CO4</b> | <b>6</b>  |
| <b>C)</b>   | Derive Dupit's equation & Explain concept of equivalent pipe.                                                                                                                                                                       | <b>CO4</b> | <b>6</b>  |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                              |            | <b>12</b> |
| <b>A)</b>   | Define a centrifugal pump. Explain the working of a single-stage centrifugal pump with sketches.                                                                                                                                    | <b>CO5</b> | <b>6</b>  |
| <b>B)</b>   | What is priming? Why is it necessary?                                                                                                                                                                                               | <b>CO5</b> | <b>6</b>  |
| <b>C)</b>   | Write a short note on multistage centrifugal pump                                                                                                                                                                                   | <b>CO5</b> | <b>6</b>  |

**\*\*\* End \*\*\***

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------|
| <p align="center"><b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b></p> <p align="center"><b>Regular &amp; Supplementary Winter Examination-2023</b></p> <p><b>Course: B. Tech.                      Branch: Mechanical &amp; Allied Engineering</b></p> <p><b>Semester: III                          Subject Code &amp; Name: BTMC302 / Fluid Mechanics</b></p> <p><b>Max Marks: 60                          Date:04-01-24                          Duration: 3Hr.</b></p> |                                                                                                                                                                                                                           |                           |             |
| <p><b><i>Instructions to the Students:</i></b></p> <p>1. All the questions are compulsory.</p> <p>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.</p> <p>3. Use of non-programmable scientific calculators is allowed.</p> <p>4. Assume suitable data wherever necessary and mention it clearly.</p>                                                                         |                                                                                                                                                                                                                           |                           |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                           | (Level/CO)                | Mark<br>s   |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Solve Any Two of the following.</b>                                                                                                                                                                                    |                           | <b>(12)</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Describe any three terms:</b><br><b>Pascal's Law,</b><br><b>Hydrostatic Law,</b><br><b>Total pressure,</b><br><b>Center of pressure</b>                                                                                | <b>(Understand / CO1)</b> | <b>(6)</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>A plate 0.125 mm distant from a fixed plate, moves at 60 cm/s and requires a force of 2 N per unit area i.e., 2 N/m<sup>2</sup> to maintain this speed. Determine the fluid viscosity between the plates in poise.</b> | <b>(Apply / CO1)</b>      | <b>(6)</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Explain the phenomenon of capillarity. Obtain the expression for capillary rise of a liquid in circular glass tube of diameter 'd'.</b>                                                                                | <b>(Understand / CO1)</b> | <b>(6)</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                           |                           |             |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Solve Any Two of the following.</b>                                                                                                                                                                                    |                           | <b>(12)</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Derive the Bernoulli's equation with the help of Euler's equation. Also discuss the assumptions made in Bernoulli's equation.</b>                                                                                      | <b>(Apply / CO2)</b>      | <b>(6)</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Define following types of flows (Any three).</b><br><b>Steady and unsteady flow</b><br><b>Uniform and non-uniform flow</b><br><b>Laminar and Turbulent flow</b><br><b>Compressible and incompressible flow</b>         | <b>(Understand / CO2)</b> | <b>(6)</b>  |

|      |                                                                                                                                                                                                                                                                                                                                                             |                    |      |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|
| C)   | An orifice meter with orifice diameter 10 cm is inserted in a pipe of 20 cm diameter. The pressure gauges fitted upstream and downstream of the orifice meter gives readings of 19.62 N/cm <sup>2</sup> and 9.81 N/cm <sup>2</sup> respectively. Co-efficient of discharge for the orifice meter is given as 0.6. Find the discharge of water through pipe. | (Apply / CO2)      | (6)  |
|      |                                                                                                                                                                                                                                                                                                                                                             |                    |      |
| Q. 3 | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                             |                    | (12) |
| A)   | A crude oil of viscosity 0.97 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 100 mm and of length 10 m. Calculate the difference of pressure at the two ends of the pipe, if 100 kg of the oil is collected in a tank in 30 seconds.                                                                              | (Apply / CO3)      | (6)  |
| B)   | Explain major energy loss and minor energy losses in pipes.                                                                                                                                                                                                                                                                                                 | (Understand / CO3) | (6)  |
| C)   | Define and explain the following terms with the help of suitable sketch.<br>Hydraulic gradient line<br>Total energy line                                                                                                                                                                                                                                    | (Understand / CO3) | (6)  |
|      |                                                                                                                                                                                                                                                                                                                                                             |                    |      |
| Q.4  | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                             |                    | (12) |
| A)   | Differentiate between<br>Stream line body and bluff body<br>Friction drag and pressure drag                                                                                                                                                                                                                                                                 | (Understand / CO4) | (6)  |
| B)   | Describe the following term with the help of neat sketch.<br>Laminar boundary layer, turbulent boundary layer, laminar sublayer and boundary layer thickness.                                                                                                                                                                                               | (Understand / CO4) | (6)  |
| C)   | What is drag and lift? Explain types of drags.                                                                                                                                                                                                                                                                                                              | (Understand / CO4) | (6)  |
|      |                                                                                                                                                                                                                                                                                                                                                             |                    |      |
| Q. 5 | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                             |                    | (12) |
| A)   | The resisting force $R$ of a supersonic plane during the flight can be considered as dependent upon the length of the aircraft $l$ , velocity $v$ , air viscosity $\mu$ , air density $\rho$ and bulk modulus of air $K$ . Express the functional relationship between these variables and the resulting force.                                             | (Apply / CO5)      | (6)  |

|           |                                                                                                       |                           |            |
|-----------|-------------------------------------------------------------------------------------------------------|---------------------------|------------|
| <b>B)</b> | <b>What is dimensionless number? Define any four dimensionless numbers.</b>                           | <b>(Understand / CO5)</b> | <b>(6)</b> |
| <b>C)</b> | <b>What is repeating variable? How are the repeating variables selected for dimensional analysis?</b> | <b>(Understand / CO5)</b> | <b>(6)</b> |
|           | <b>*** End ***</b>                                                                                    |                           |            |

**The grid and the borders of the table will be hidden before final printing.**